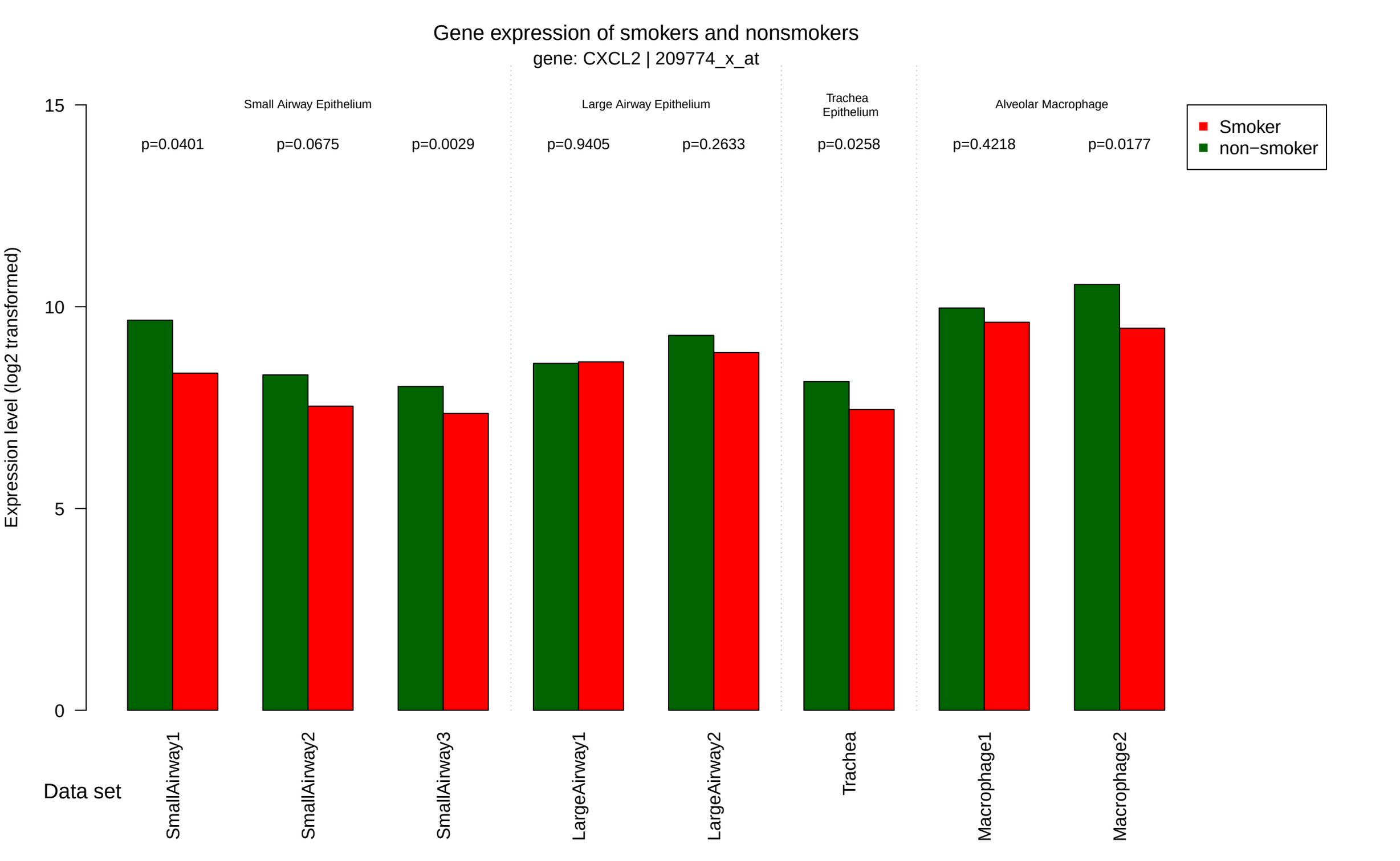
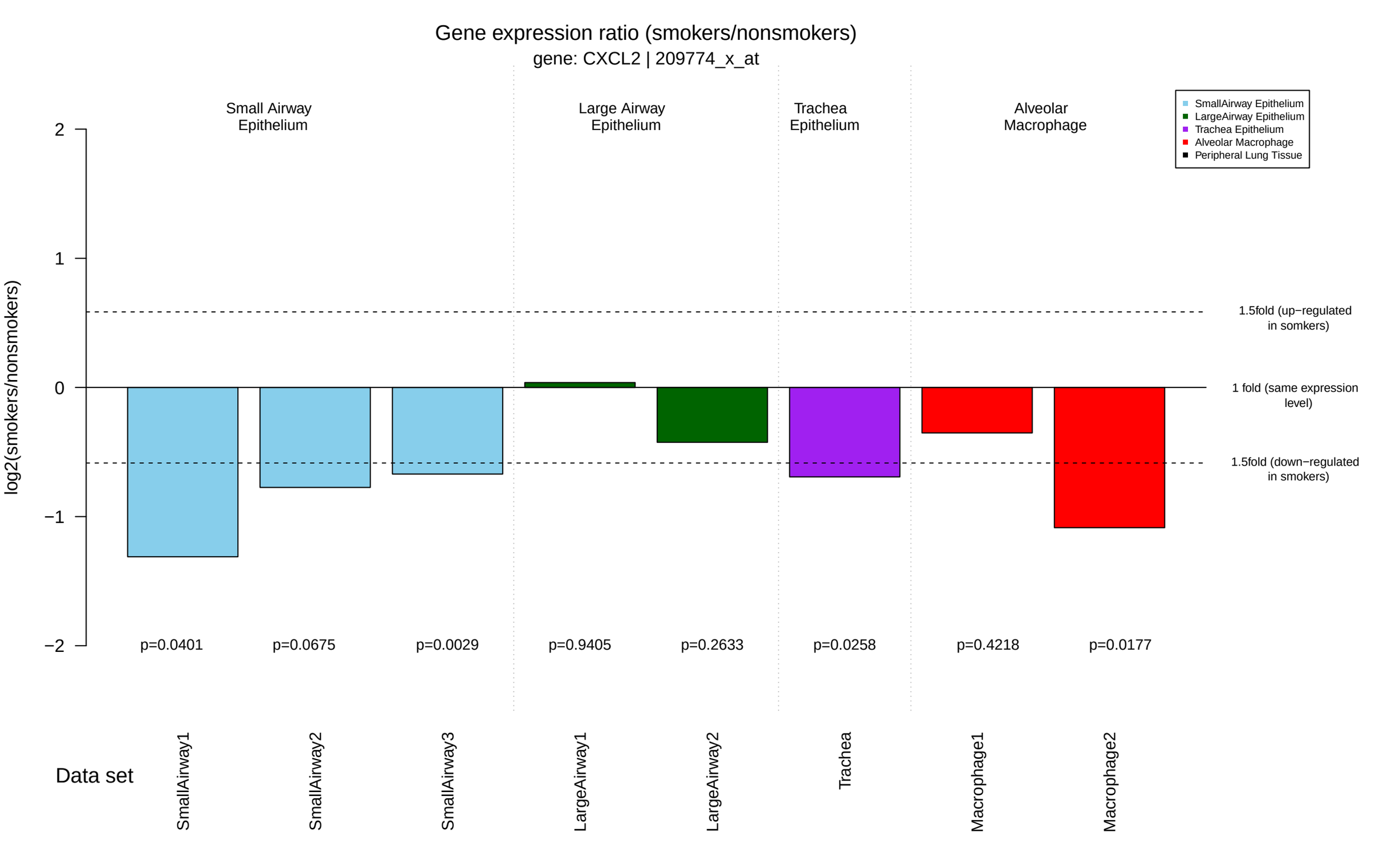
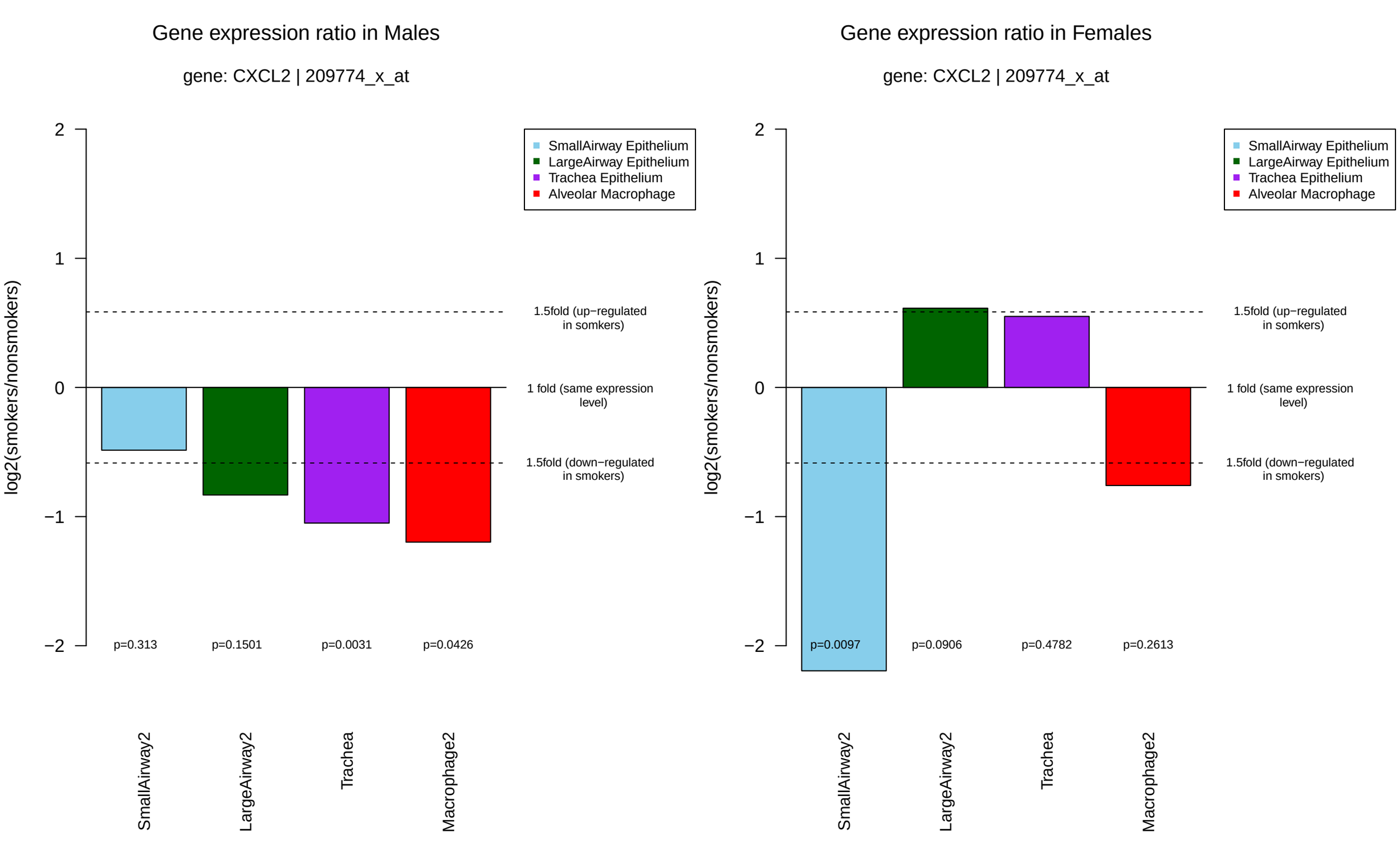
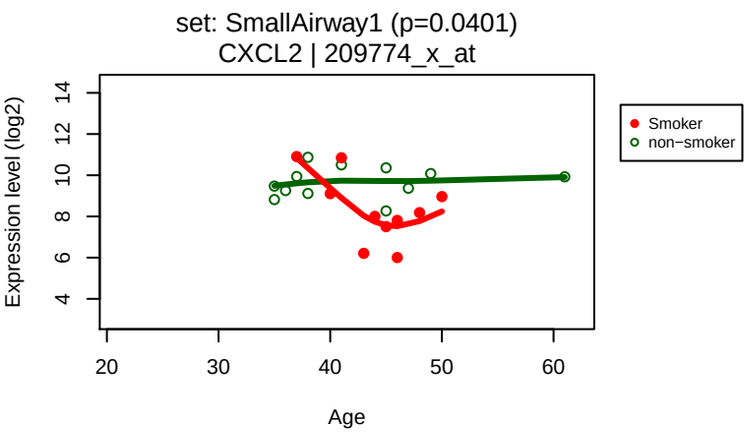
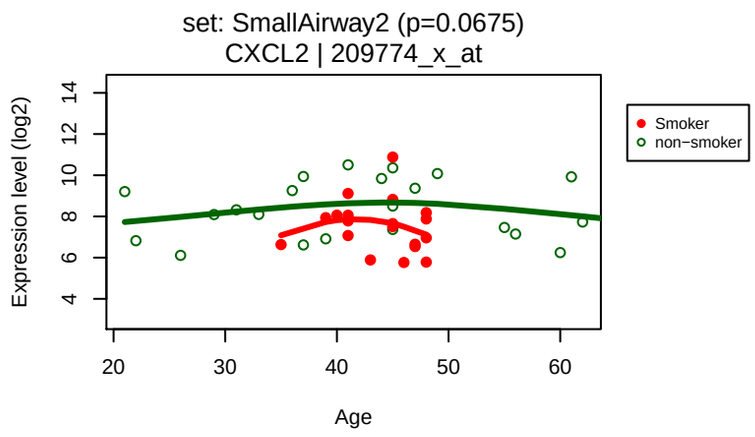
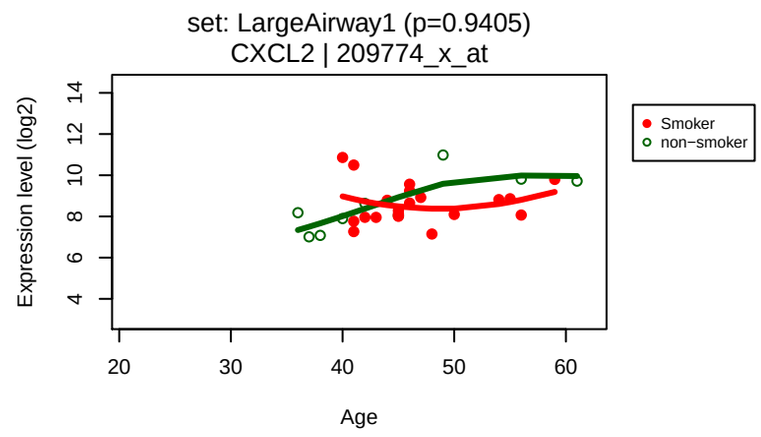

Gene expression of smokers and nonsmokers
gene: CXCL2 | 209774_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0401
p=0.0675
p=0.0029
p=0.9405
p=0.2633
p=0.0258
p=0.4218
p=0.0177
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

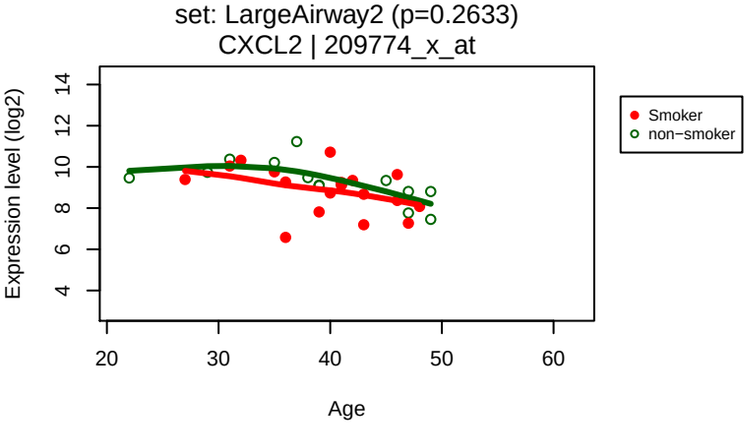

Gene expression ratio (smokers/nonsmokers)
gene: CXCL2 | 209774_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0401
p=0.0675
p=0.0029
p=0.9405
p=0.2633
p=0.0258
p=0.4218
p=0.0177
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

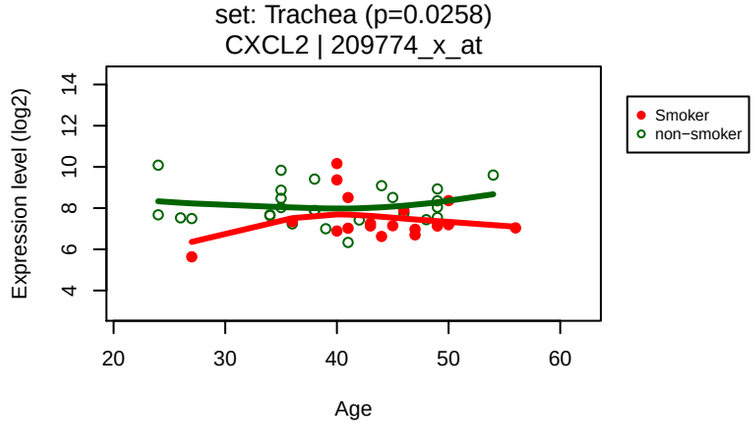

Gene expression ratio in Males
gene: CXCL2 | 209774_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.313
p=0.1501
p=0.0031
p=0.0426
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CXCL2 | 209774_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0097
p=0.0906
p=0.4782
p=0.2613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

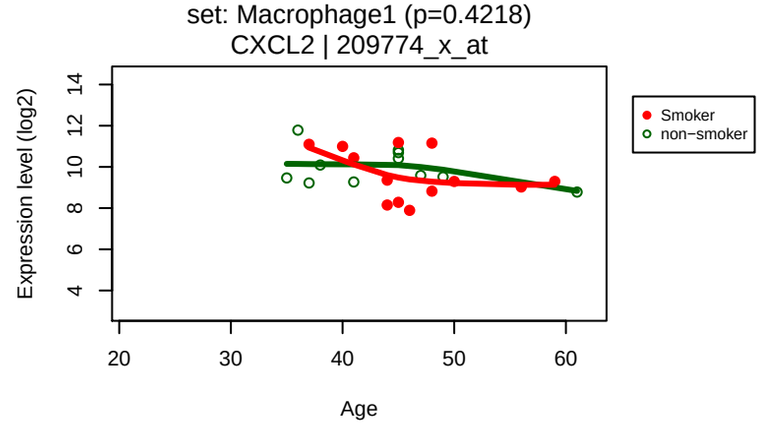

set: SmallAirway1 (p=0.0401)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker

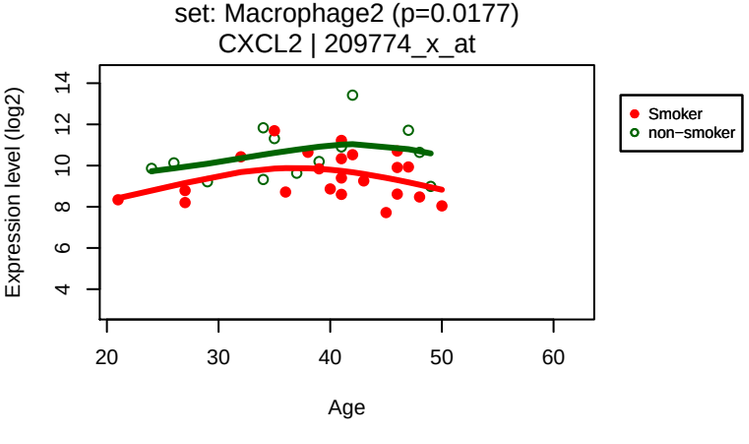

set: SmallAirway2 (p=0.0675)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker

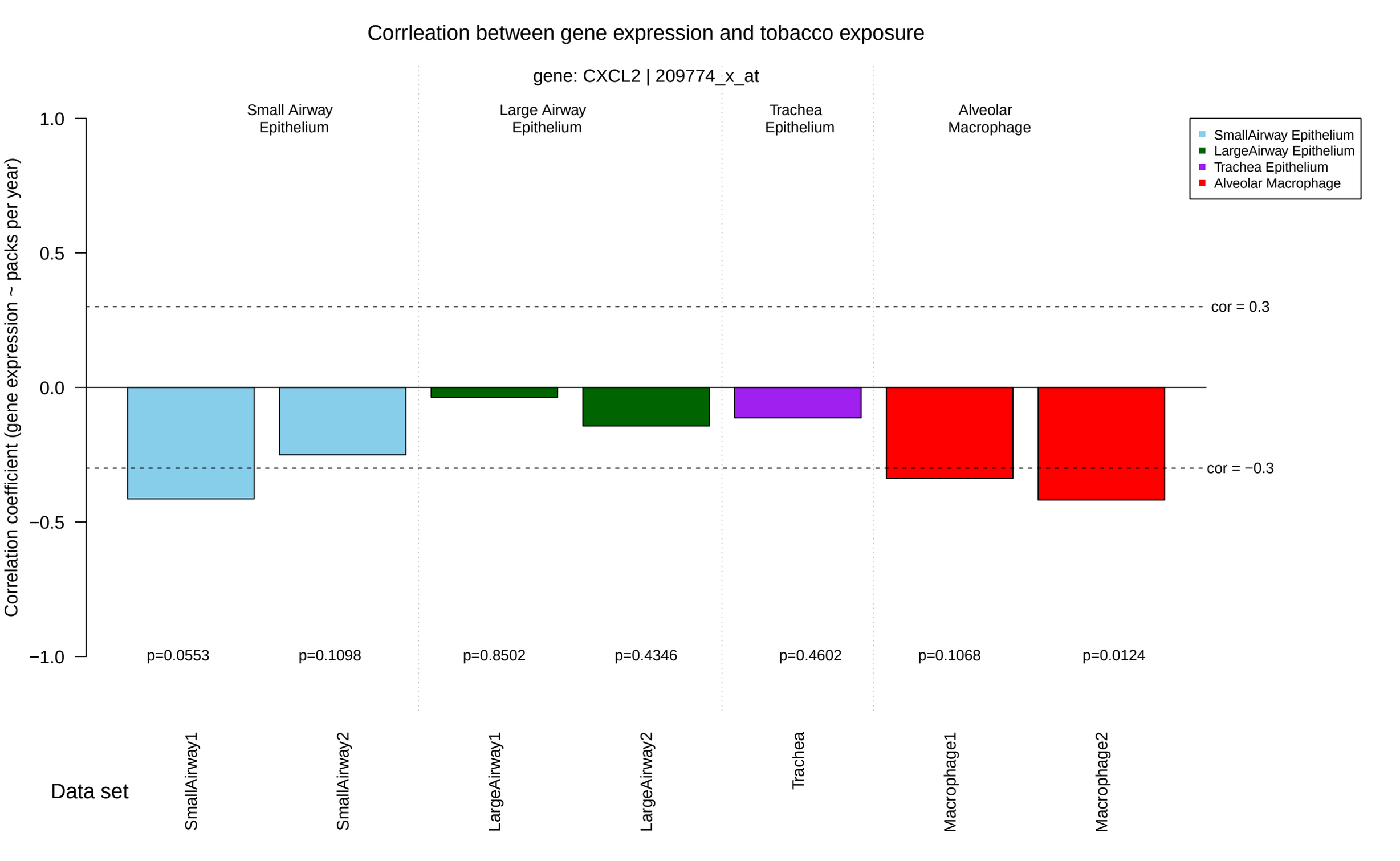

set: LargeAirway1 (p=0.9405)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2633)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0258)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4218)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0177)
CXCL2 | 209774_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CXCL2 | 209774_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0553
p=0.1098
p=0.8502
p=0.4346
p=0.4602
p=0.1068
p=0.0124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2